



Qubit Control Primer

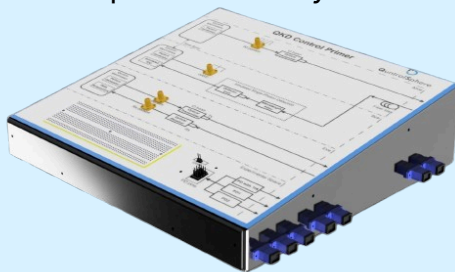
QKD Networks Primer

The QKD networks primer introduces the operational dynamics of Quantum Key Distribution (QKD) implementation and its control with state-of-the-art features like:

- **QKD emulation platform**
 - Hands-on hardware experience on Coherent-One Way (CoW) QKD protocol.
 - QKD fundamentals under different test environments like laser power variation, modulation patterns, noise sources, fibre types, TIA design, blinding attack, etc.
- **QKD network digital twin**
 - Hands-on towards classical-QKD network integration, key access mechanism, QKD applications and use cases development, networking control and monitoring etc.

Learning outcomes

- Fundamentals of elementary blocks of a fibre-based QKD system.
- QKD experiments under different test conditions, including attacks.
- Integration of QKD with the classical network.
- QKD application development and key access mechanism



About QuntrolSphere:

QuntrolSphere, associated with IIT Madras' startup ecosystem, aims to be a leader in distributed quantum computing technology stack, with specific focus on design and development of control systems for quantum technologies, transduction technology, optical interconnects and quantum networking.

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Quantum Computing Primer

The quantum computing primer enables hands-on experience towards niche aspects of quantum computing and control systems with state-of-the-art features like:

- **Multi-channel Control Electronics platform**
 - Hands-on experience on superconducting qubit RF control and readout.
 - Focus on spectrum analysis, phase noise, SFDR, sampling, aliasing etc.
- **Superconducting qubit emulator**
 - Hands-on experience in interfacing control electronics with QPU.
 - Perform QPU characterisation experiments like cavity spectroscopy, qubit spectroscopy, T1/T2 measurements and more.
- **Low cost programmable arbitrary waveform generator and data acquisition board**
 - Hands-on hardware and software development towards superconducting qubit control and readout.

Learning outcomes

- Fundamental skills towards control electronics development and characterisation.
- Analog and digital circuits, measurement and instrumentation.
- Control electronics integration with QPU and qubit characterisation.
- Arbitrary waveform generation and data acquisition on FPGA and their characterisation.

